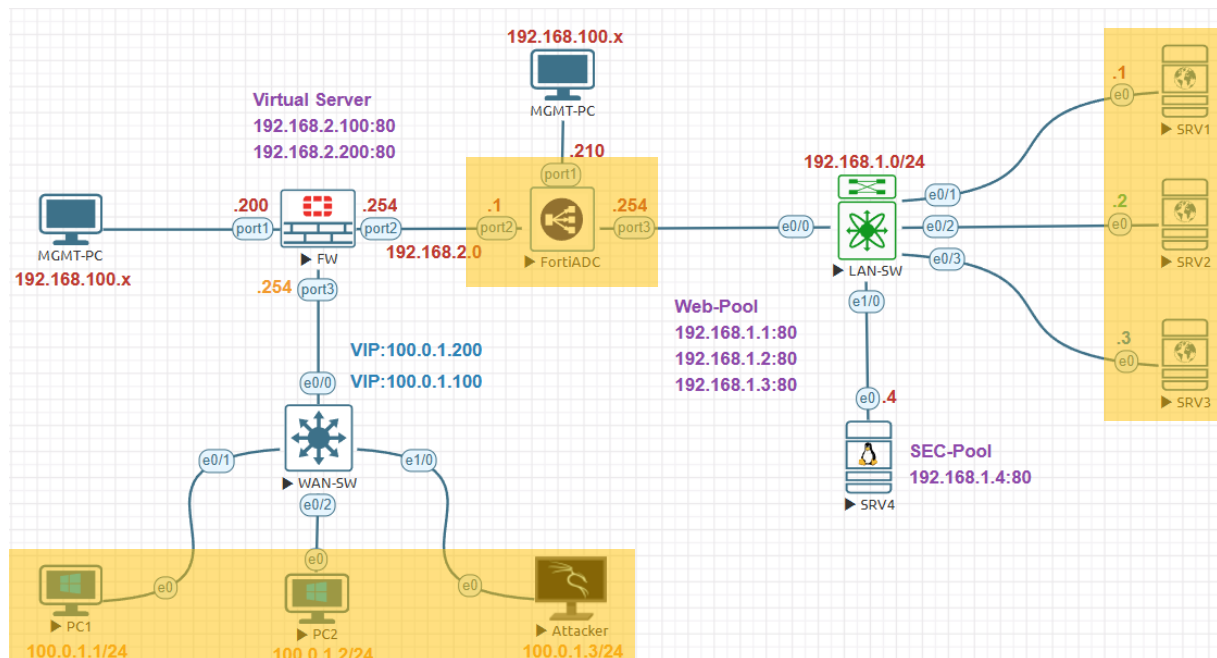


Intrusion Prevention Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Click **Network Security > Intrusion Prevention**. There is pre-defined Intrusion Prevention profile OR we can create our own in this case we will use **protect_http_server** predefined profile

The screenshot shows the FortiGate Web UI. On the left, the 'Network Security' menu is expanded, and 'Intrusion Prevention' is selected. On the right, a table lists predefined intrusion prevention profiles. The 'protect_http_server' profile is highlighted with a red box.

Name	Comments
default	Prevent critical attacks.
all_default	All predefined signatures with default setting.
sniffer-profile	Monitor IPS attacks.
all_default_pass	All predefined signatures with PASS action.
protect_client	Protect against client-side vulnerabilities.
protect_http_server	Protect against HTTP server-side vulnerabilities.
protect_email_server	Protect against email server-side vulnerabilities.
high_security	Blocks all Critical/High/Medium and some Low severity
IPS-Profile	

Click **Server Load Balance > Virtual Server > Virtual Server**. Click the edit icon or double click to edit the Layer 4 virtual server **WEB-VS**.

The screenshot shows the FortiGate Web UI for the Virtual Server configuration page. The 'Virtual Server' tab is selected. A table lists three virtual servers. The 'WEB-VS' entry is highlighted with a red box.

Name	Type	Address	Port	Profile
WEB-VS	Layer 4	192.168.2.100	80	LB_PROF_TCP
SEC-VS	Layer 7	192.168.2.200	80	LB_PROF_HTTP
HTTPS-VS	Layer 7	192.168.2.100	443	HTTPS_Profile

The screenshot shows the FortiGate Web UI for the configuration of the 'WEB-VS' virtual server. The 'Basic' tab is selected. The configuration details are as follows:

- Name: WEB-VS
- Type: Layer 4 (selected)
- Status: Enable
- Address Type: IPv4
- Traffic Group: default
- Comments: Specify the comments

On the **Security** tab, change the **IPS Profile** setting to **protect_http_server**, and then click **Save**.

Virtual Server

Basic General **Security** Monitoring

DoS Protection Profile Click to select

IPS Profile protect_http_server

Server Load Balance

Virtual Server

Virtual Server

Basic General **Security** Monitoring

DoS Protection Profile Click to select

IPS Profile protect_http_server

Save Cancel

IPS profile only applied to type Layer 4 virtual Sever only.

Virtual Server

Content Rewriting

Content Routing

NAT Source Pool

Schedule Pool

Delete

+ Create New

Search

Add Filter

Name	Type	Address	Port	Profile
WEB-VS	Layer 4	192.168.2.100	80	LB_PROF_TCP
SEC-VS	Layer 7	192.168.2.200	80	LB_PROF_HTTP
HTTPS-VS	Layer 7	192.168.2.100	443	HTTPS_Profile

Showing 1 to 3 of 3 entries

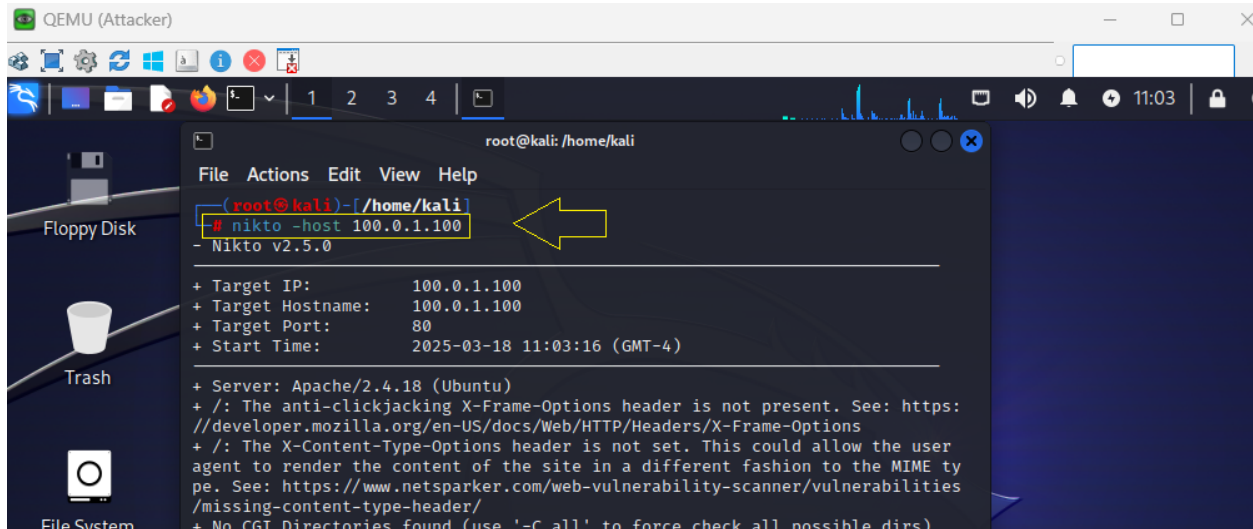
0 rows selected

Show 25 entries

Testing & Verification:

On the External Kali Linux, open Terminal type below command:

Nikto -host 100.0.1.100



```
root@kali: /home/kali
File Actions Edit View Help
(root@kali)-[/home/kali]
# nikto -host 100.0.1.100
- Nikto v2.5.0

+ Target IP: 100.0.1.100
+ Target Hostname: 100.0.1.100
+ Target Port: 80
+ Start Time: 2025-03-18 11:03:16 (GMT-4)

+ Server: Apache/2.4.18 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
```

Navigate to **Log & Report > Security Log** > choose from dropdown **IPS** to verify the logs

Dashboard

>

Security Fabric

>

FortiView

>

System

>

Shared Resources

>

Network

>

Server Load Balance

>

Link Load Balance

>

Global Load Balance

>

Web Application Firewall

>

Network Security

>

DoS Protection

>

User Authentication

>

Log & Report

>

Traffic Log

>

Security Log

>

Download

Refresh

IPS

25/03/07 21:19:19 - 25/03/18 18:04:02

Add Filter

Date	Time	Source	Destination	Service
2025-03-18	18:03:56	100.0.1.3	192.168.2.100	tcp
2025-03-18	18:03:46	100.0.1.3	192.168.2.100	tcp
2025-03-18	18:03:36	100.0.1.3	192.168.2.100	tcp
2025-03-18	18:03:26	100.0.1.3	192.168.2.100	tcp
2025-03-18	18:03:19	100.0.1.3	192.168.2.100	tcp
2025-03-18	18:03:19	100.0.1.3	192.168.2.100	tcp
2025-03-18	17:26:52	100.0.1.3	192.168.2.100	tcp
2025-03-18	17:26:45	100.0.1.3	192.168.2.100	tcp
2025-03-18	17:26:35	100.0.1.3	192.168.2.100	tcp
2025-03-18	17:26:25	100.0.1.3	192.168.2.100	tcp
2025-03-18	17:26:15	100.0.1.3	192.168.2.100	tcp

Download

Refresh

IPS

25/03/07 21:19:19 - 25/03/18 18:04:02

Add Filter

Date	Time	Source	Destination	Service	Severity	Rule Name
2025-03-18	18:03:56	 100.0.1.3	192.168.2.100	tcp	critical	Bash.Function.Definitions.Remote.Code.Execution
2025-03-18	18:03:46	 100.0.1.3	192.168.2.100	tcp	critical	Bash.Function.Definitions.Remote.Code.Execution
2025-03-18	18:03:36	 100.0.1.3	192.168.2.100	tcp	critical	Bash.Function.Definitions.Remote.Code.Execution
2025-03-18	18:03:26	 100.0.1.3	192.168.2.100	tcp	critical	Bash.Function.Definitions.Remote.Code.Execution
2025-03-18	18:03:19	 100.0.1.3	192.168.2.100	tcp	critical	Bash.Function.Definitions.Remote.Code.Execution
2025-03-18	18:03:19	 100.0.1.3	192.168.2.100	tcp	medium	Apache.Expect.Header.XSS
2025-03-18	17:26:52	 100.0.1.3	192.168.2.100	tcp	critical	Bash.Function.Definitions.Remote.Code.Execution
2025-03-18	17:26:45	 100.0.1.3	192.168.2.100	tcp	critical	Bash.Function.Definitions.Remote.Code.Execution
2025-03-18	17:26:35	 100.0.1.3	192.168.2.100	tcp	critical	Bash.Function.Definitions.Remote.Code.Execution